

8. Cerebellum (functions)

Areas of cerebellum (figure 34 and table 14)

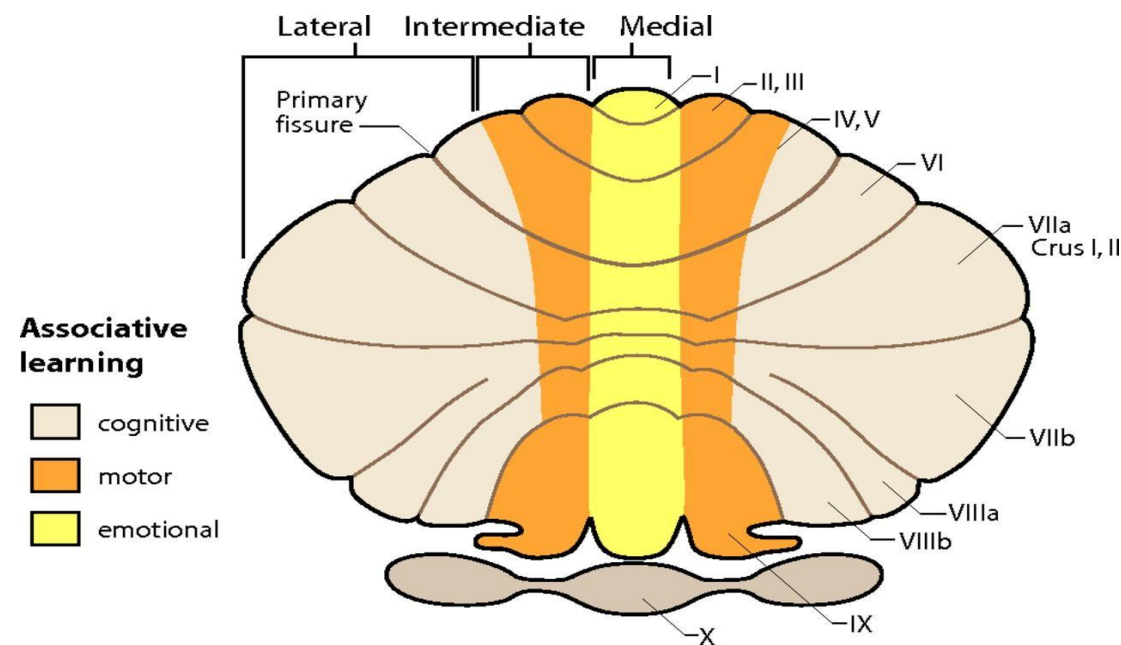


Figure 34 The cerebellum

Table 14 : Functional Areas of Cerebellum

REGION	FUNCTIONS	MOTOR PATHWAYS INFLUENCED
Lateral hemispheres	Motor planning for extremities	Lateral corticospinal tract
Intermediate hemispheres	Distal limb coordination	Lateral corticospinal tract, rubrospinal tract
Vermis and flocculonodular lobe	Proximal limb and trunk coordination	Anterior corticospinal tract, reticulospinal tract, vestibulospinal tract, tectospinal tract
	Balance and vestibulo-ocular reflexes	Medial longitudinal fasciculus

Vestibulocerebellum

- is composed of *flocculondular lobe* and the portion of the vermis immediately adjacent to the flocculondular lobe.
- provides neural circuits for most of the body's equilibrium movements

Role:

1. Function in Association with the Brain Stem and Spinal Cord to Control Equilibrium and Postural Movements.
2. calculate in advance rates and directions of movement , where the different parts will be during the next few milliseconds thus control the progression to the next sequential movement.
3. provide anticipatory correction of postural motor signals necessary for maintaining equilibrium even during extremely rapid motion, including rapidly changing directions of motion.

Spinocerebellum includes

1. both the vermis and middle parts of the cerebellar hemisphere, or paravermis.
2. provides the circuitry for coordinating mainly movements of the distal portions of the limbs, especially the hands and fingers.

Role:

1. provides smooth, coordinate movements of the agonist and antagonist muscles of the distal limbs for performing acute purposeful patterned movements
2. Prevent Overshoot of Movements and to “Damp” Movements.
3. Control of Ballistic Movements

Cerebrocerebellum:

formed from lateral hemispheres of the cerebellum

Critical for *planning movement* and evaluating sensory information required for action.

Plays critical role for planning anticipated motor movements and modulating at least some cognitive functions.

receives virtually all its input from the cerebral motor cortex and adjacent premotor and somatosensory cortices of the cerebrum.

transmits its output information in the upward direction back to the brain, functioning in a feedback manner with the cerebral cortical sensorimotor system to plan sequential voluntary body and limb movements, planning these as much as tenths of a second in advance of the actual movements. This is called development of “motor imagery” of movements to be performed

Role:

1. ability to progress smoothly from one movement to the next in orderly Succession
2. to provide appropriate timing for each succeeding movement
3. provides a “time-base,” perhaps using time-delay circuits, against which signals from other parts of the central nervous system can be compared

FUNCTIONS OF CEREBELLUM

1. Role in control voluntary Movement (figure 35)

- Improves *accuracy* of movements by comparing descending motor commands with information regarding current motor actions.

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- Send *corrections* as necessary to equate actual with desired motor sequences by acting on the brain stem and cortical motor areas.
1. Cerebellum receives input from most body senses and serves as distinct *feed-back loop* for sensory-motor coordination
 2. Without feedback, poor or no coordinated movement
 3. Cerebellum can be said to compare *what* the body is doing to *what should* the body be doing and signals the brain to make changes as necessary.
 4. Damage to the cerebellum typically results in
 - loss of ability for both sensory and motor coordination
 - Results in disruptions of equilibrium, motor coordination and postural control.

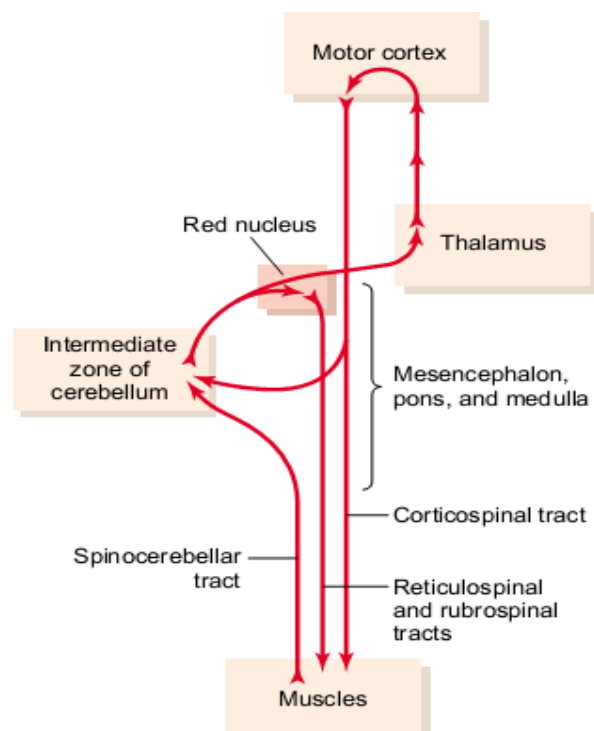


Figure 35: circuit for cerebellar control of voluntary movements

2. Role in Timing and programming of voluntary movement

- Determine future position of the moving limb and control the rate, range, force, and direction of movement during rapid progressive and sequential motor movement such as writing, walking, running and talking.
- Mediated with the help of two way neuronal connections between cerebellum and premotor & sensory cortex.

3. Role in Involuntary Movement

- Control the rate of firing level of spinal neurons through various pathways of medial system to cause co-ordinated and precise reflex contraction of muscle

4. Damping function

- Without cerebellum movements are jerky and pendular.
- Appropriate learned, subconscious signals stop the movement precisely at the intended point, thereby preventing the overshoot as well as the tremor.

a) Comparator and Error- Correction Mechanism

- The intermediate zone of the spinocerebellum essentially acts as a "comparator" that compares the motor intentions of the higher centers with the actual performance of the involved ms.
- When there is any "error" in performance or "deviation" from the original plan of the intended voluntary motor act, then the intermediate zone and the interposed nucleus send 'corrective signals' back to the

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motor areas of the CC and the red nucleus, which give origin to the descending motor tracts innervating mainly the lower motor neurons of the distal limb ms

b) Predictive and Damping Mechanism

- The cerebellum receives information regarding the velocity and direction of the intended movement.
- The cerebellum would predict from these informations how far that part of the body will move in a given time, and uses this information to determine the precise time to damp the movement, and then it sends its decision to the motor cortex to stop the ongoing movement exactly at the intended position.

Planning the Sequence and Timing of Movements

c) Planning the Sequence and Timing of Movements

i. Planning the Sequence of Movements

- The cerebrocerebellum uses the information provided from the CC and the BG for planning the sequence of contraction of the different ms involved in the voluntary motor act, to achieve the goal of the movement.
- Then, the "plan" of the movement sequence is transmitted from the cerebrocerebellum to the motor areas of the CC, where it is used to adjust the final motor command before it is discharged to the lower motor centers.

ii. Timing of Movements

- Also the cerebrocerebellum is to provide perfect timing of voluntary movements.

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- This is established by computing (calculating) the appropriate timing for the "onset" and "termination" of contraction of each of the ms involved in the performance of the successive movements during voluntary motor acts → assures the smooth progression of the whole movement.

5. Control of Equilibrium and Postural Movements

- Controlling balance between agonist and antagonist muscle contractions of the spine, hips, and shoulders during rapid changes in body positions based on the information from both the body periphery and the vestibular apparatus is used in a typical feedback control circuit.
- Provide anticipatory correction of postural motor signals necessary for maintaining equilibrium even during extremely rapid motion, including rapidly changing directions of motion.

When the equilibrium is disturbed or exposed to acceleration → ++ the vestibular receptors → send sensory signals to Vestibulocerebellum which initiate immediate corrective signals that are sent to:-

- i) The vestibular nuclei, and RF → adjust the tone and contractility of the axial and proximal limb ms. This helps to maintain equilibrium during the change in head position, and during exposure to acceleration or active movements of the body.
 - ii) The superior colliculus and the medial longitudinal bundle → to coordinate eye movements with head movements during exposure to acceleration → to maintain clear vision which is important for keeping equilibrium during head movements.
- The vermis is the principal region of the cerebellum concerned with

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postural adjustment.

- It receives sensory information from ms and joint proprioceptors (particularly from the axial regions), concerning "position" of the body.
- Its output controls the vestibulospinal and reticulospinal tracts that regulate the tone and contraction of the axial and proximal limb ms .

CEREBELLAR SYNDROME

CLINICAL FEATURES OF CEREBELLAR SYNDROME

1. Cerebellar lesion usually must involve one or more of the deep cerebellar nuclei—the dentate, interposed, or fastigial nuclei
2. Clinical features includes the triad:
 - A. Hypotonia
 - Decreased tone of the peripheral body musculature on the side of the cerebellar lesion.
 - Due to Loss of the deep cerebellar nuclei, particularly of the dentate and interposed nuclei
 - Results in to loss of cerebellar facilitation of the motor cortex and brain stem motor nuclei by tonic signals from the deep cerebellar nuclei.
 - B. Asthenia
 - There is weakness of movements and the involved ms fatigue more readily than do normal ms, resulting from interruption of the activating effect of the cerebellum on the cerebral cortical motor areas.
 - C. Ataxia (figure 36).
 - One of the most important symptoms of cerebellar disease is instability

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when standing/walking.

- Result from lesions in the spinocerebellar tracts because feedback information from the moving parts of the body to the cerebellum is essential for cerebellar timing of movement termination.

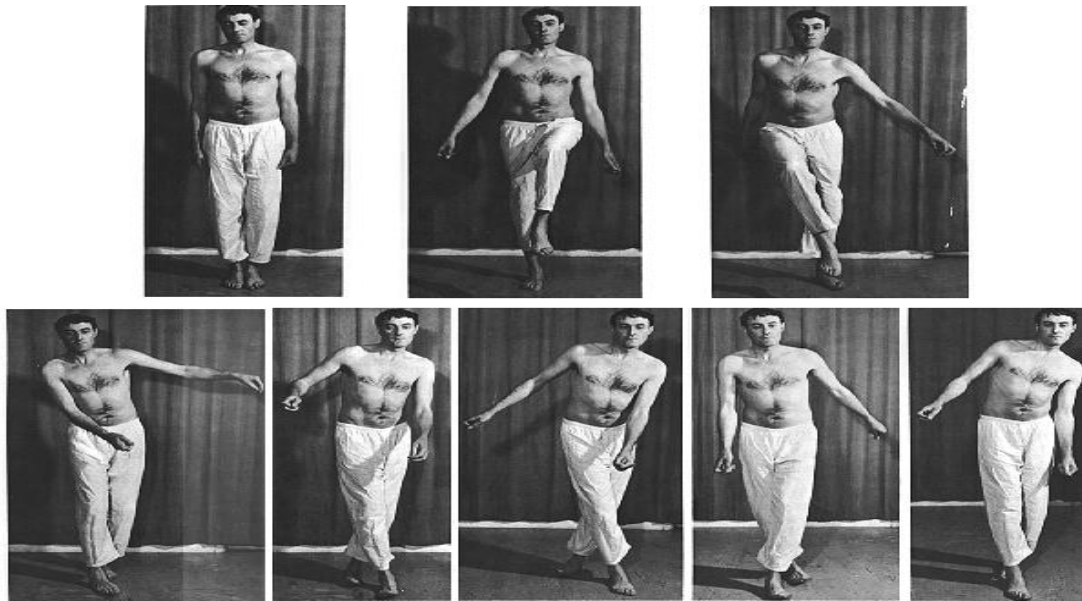


Figure 36: Cerebellar Ataxia

3. Ataxia manifestations are:

i. Dysmetria (figure 37)

- ✓ Inability to judge distance and when to stop.
- ✓ Loss of ability to turn off the movement at intended mark.
- ✓ also called as past pointing, means a person ordinarily moves the hand or some other moving part of the body considerably beyond the point of intention.
- ✓ results from the fact that normally the cerebellum initiates most of the motor signal that turns off a movement after it is begun; if the cerebellum is not available to do this, the movement ordinarily goes beyond the intended mark



Figure 37: Dysmetria

ii. Intentional Tremor

- ☐ The movements tend to oscillate, especially when they approach the intended mark, first overshooting the mark and then vibrating back and forth several times before settling on the mark.
- ☐ Results from cerebellar overshooting and failure of the cerebellar system to “damp” the motor movements.

iii. Decomposition of complex movement

- ☒ The motor act is carried out as several fragmented steps rather than a smoothly progressing movement.
- ☒ For instance, in reaching for an object by the hand, the cerebellar patient may first move the shoulder joint, then the elbow, followed by the wrist and fingers → simulate movements of a "robot".

iv. Rebound phenomenon.

- The cerebellar patient is unable to stop the ongoing movement rapidly due to failure of the predictive and damping functions of the cerebellum. This can be observed in what is called "rebound phenomenon".
- When there is a flexion of the forearm against resistance (provided by the examiner's hand), the cerebellar patient cannot stop the

resultant inward movement of his limb in due time following its release, and the forearm flexes forcibly and may strike his body with considerable violence.

v. Dysdiadochokinesia

- inability to perform rapid alternating movements
- Loss of ability to arrest one motor impulse and substitute the opposite, e.g. pronate and supinate forearm + hand quickly
- motor control system fails to predict position of different parts of the body at a given time during rapid motor movements. Leading to incoordinate progression of movement i.e. movements overshoot, undershoot, irregular or inaccurate

vi. Cerebellar Nystagmus

- Cerebellar nystagmus is tremor of the eyeballs results in rapid, tremulous movements of the eyes rather than steady fixation
- usually occurs usually when one attempts to fix the eyes on a scene to one side of the head.
- manifestation of failure of damping by the cerebellum.
- occurs due to damage of the flocculonodular lobes of the cerebellum
- also associated with loss of equilibrium because of dysfunction of the pathways through the flocculonodular cerebellum from the semicircular ducts.

vii. Dysarthria

- Inability to adjust in advance intensity of sound & duration of each successive sound leading to jumbled vocalization, with some syllables

loud, some weak, some held for long intervals, some held for short intervals, and resultant speech that is often unintelligible.

- Failure of progression occurs in talking because the formation of words depends on rapid and orderly succession of individual muscle movements in the larynx, mouth, and respiratory system.

viii. Unsteady of gait.

- The gait is unsteady and broad-based due to dysmetria and kinetic tremors of the lower limbs.

9. Functions of vestibular apparatus

The maintenance of the erect posture is mediated through two antagonistic forces:

- The force of gravity.
- The tonic contraction of antigravity muscles.
- The center of gravity changes continuously according to body posture.
- The postural reflexes increase the tone of antigravity muscles to maintain the erect posture each time the body position changes.
- The body has no receptors for constant speed motion. We do not feel the constant speed movement of earth at the high speed of 30 Km/second.
- The body can detect:
 - A. Movement with changing rate (acceleration).
 - B. Movement of changing direction which may be: linear or angular.